ReRAM based processing engine: ReSiPE

In the last two years we have moved to single spiking designs. The goal is to leverage the timing of the spike for data representation and the matrix operation. For that we divide the design into three stages.

In the first stage we translate the timing of the input spikes to an analogue voltage. Hence, we use a charging process of a capacitor Cgd for reference. When the input spikes 'S' then the voltage will be sampled at exactly the spiking time.

In stage two, the hinge signals are fed into the ReRAM crossbar arrays to perform matrix operation and during the computation the capacitor Ccog is charged continuously. The charging speed is determined by the bit-line current.

The last stage is the output stage where Vout is translated to upper spike timing. The voltage of Cgd is again used as the reference and it generates the Vout here (see Figure 11).

There are three important components that are needed while building machine learning techniques such as data, model, and application. All these elements are connected and operate together. Although we can say that a system is smart, it can still be attacked, hence, further creating a lot of security problems.

Many researchers have shown how well-trained deep learning algorithms can be fooled with adversary tags intentionally. Most of the work focuses on introducing minimal perturbation of the data, which results in a maximally devastating effect on the model. In the case of image based data, it works to disrupt the network's ability to classify with imperceptible perturbations to the images.

These attacks are of great concern because we do not have a firm understanding of the occurrences taking place inside these interpretable deep models. However, attacks may also provide a way to study the inner workings of such models. There are two types of attacks—the transfer based attacks and activation attacks.

Transfer based attack: Figure 12 shows two different models that are trained with the data from the same distribution. It implies the property of adversarial transferability. This is particularly relevant for a targeted attack. The region for a targeted class in feature space must have the same orientation with respect to the source image for the transferred target to be attacked successfully. In our set-up we train a white box model or source model on the same task as the black box or the target model.

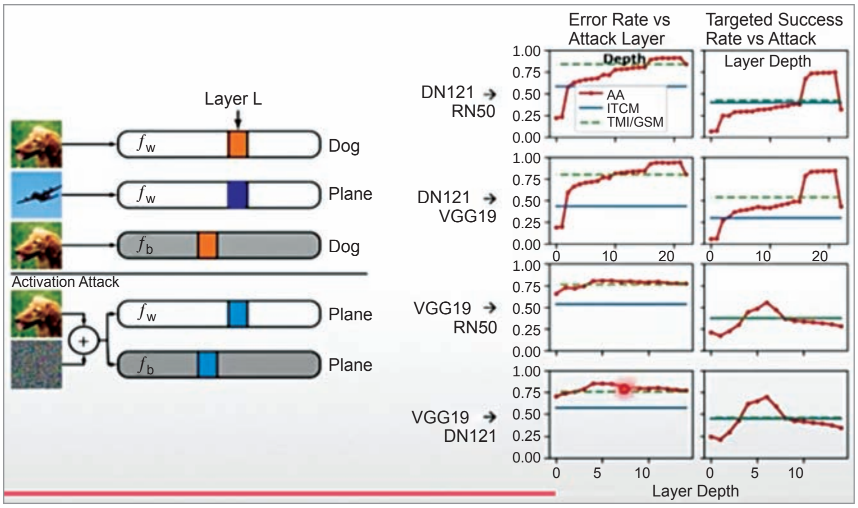

Figure 13: Activation attack

Activation attack: Here we have designed an activation attack by using the properties of adversarial transferability. As shown in Figure 13, the white box model (Fw) and a black box model (Fb) are initially correct. We perturb the source image, which drives the layer L activation of the dog image and perturbs the plane image. The results can be seen on the right hand side.

Adversarial vulnerability isolation

To minimise the attack transferability, we studied why attacks even transfer. Inspired by Eli's paper we found that models entrained on the same data set capture a similar set of non-robust features. These are the features that are highly correlated to the labels yet sensitive to noise. Such features open the door for vulnerabilities and adversary attacks.

We found that a large overlap in the non-robust features leads to overlapping vulnerabilities across different models which, in turn, leads to transferability of the attacks. For every source image and tested set, we randomly sampled another target image and non-robust images that look like the source while having the same hidden features as the target in the hidden layer of the model.

To sum it all up, it is quite understandable that the future computing systems will evolve making them more user-friendly, adaptive, and highly cost-efficient. It is a holistic scheme that integrates the efforts on device, circuit, architecture, system as well as algorithm. To achieve all of this there is a lot of work that needs to be done and a lot of challenges to be met. END

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